HI-634 Assignment 1: Option 2 — UCI Heart Disease

Predicting Coronary Artery Disease via Non-Invasive Clinical Markers

Course: HI-634 Health Informatics | Date: October 29, 2026 | Environment: Python 3.13 (.venv)

EXECUTIVE SUMMARY

This study establishes a reproducible, leakage-free clinical machine learning pipeline to predict the presence versus absence of significant coronary artery disease (>50% angiographic stenosis) using the Cleveland Clinic cohort from the UCI ML Repository. Through exploratory auditing, strict training-fold preprocessing, and rigorous comparative evaluation against a baseline, an ensemble Random Forest was identified as the optimal diagnostic decision support tool. Evaluated on an untouched hold-out test cohort (N = 61), Random Forest achieved an Accuracy of 91.8%, an ROC-AUC of 0.959, and an exceptional Sensitivity of 92.9% (26/28 cardiac cases detected), while maintaining a high Specificity of 90.9% (30/33 controls ruled out). This report stands entirely on its own as a complete methodological, empirical, and translational health informatics evaluation.

1. Question and Data Source

Focused Research Question:
"*Can routine clinical examination and non-invasive diagnostic exercise measurements predict the presence (versus absence) of significant coronary artery disease (>50% diameter stenosis) in adult cardiac patients?*"

- Problem Formulation: Formulated as a supervised binary classification task mapping 13 clinical predictors to a discrete state: target = 0 (absence: <50% stenosis) versus target = 1 (presence: >50% diameter stenosis in one or more vessels).

- Dataset Citation & Official URL: Data was retrieved from the UCI Machine Learning Repository: Heart Disease Database (Cleveland Clinic Foundation, ID 45). URL: https://archive.ics.uci.edu/dataset/45/heart+disease. Citation: Detrano et al. (1989), American Journal of Cardiology. Protocol strictly isolates the single-center Cleveland Clinic cohort (N = 303).

- Unit of Observation & Clinical Significance: An individual adult patient clinical encounter (N = 303 unique patient records). Each record represents an independent patient encounter without longitudinal clustering. Diagnostic sensitivity and specificity directly measure individual triage safety at the primary-to-specialty cardiology referral boundary.

- Rationale for Data Suitability: The Cleveland database is uniquely matched to this inquiry because coronary angiography (the historical gold standard) was systematically performed on every enrolled patient, establishing objective ground truth. The predictor set captures the standard diagnostic cascade: patient history (age, sex, chest pain), resting vitals (blood pressure, cholesterol, resting ECG), and functional provocation markers (peak heart rate, exercise angina, ST depression, ST slope, fluoroscopy, thallium).

2. Initial Exploration & Data Quality Audit

A structural audit was conducted on data/raw/processed.cleveland.data. The raw file is an ASCII headerless CSV of exactly 303 rows and 14 columns, representing missing entries with '?'. Duplicate row analysis confirmed 0 duplicate rows (100% unique encounte

Clinical Data Dictionary (The 14 Documented Attributes)

#	Attribute	Type & Units	Clinical Meaning & Codebook Definition
1	age	Continuous (Yrs)	Chronological age (range: 29–77 yrs; mean 54.4 ± 9.0)
2	sex	Binary (0, 1)	Biological sex: 1 = Male (68.0%), 0 = Female (32.0%)
3	cp	Nominal (1..4)	Chest pain: 1: Typical (7.6%), 2: Atypical (16.5%), 3: Non-anginal (28.4%), 4: Asymptomatic (47.5%)
4	trestbps	Continuous (mm Hg)	Resting blood pressure on hospital admission (range: 94–200 mm Hg; median 130.0)
5	chol	Continuous (mg/dl)	Serum cholesterol (range: 126–564 mg/dl; median 241.0; severe right-tail skewness)
6	fbs	Binary (0, 1)	Fasting blood sugar > 120 mg/dl: 1 = True (14.9%), 0 = False (85.1%)
7	restecg	Nominal (0..2)	Resting ECG: 0: Normal (49.8%), 1: ST-T wave abn (1.3%), 2: Left ventr hypertrophy (48.8%)
8	thalach	Continuous (bpm)	Max heart rate achieved during treadmill stress test (range: 71–202 bpm; median 153.0)
9	exang	Binary (0, 1)	Exercise-induced angina: 1 = Yes (32.7%), 0 = No (67.3%)
10	oldpeak	Continuous (mm)	ST depression induced by exercise relative to rest (range: 0.0–6.2 mm; median 0.8)
11	slope	Ordinal (1..3)	Slope of peak exercise ST segment: 1: Upsloping (46.9%), 2: Flat (46.2%), 3: Downsloping (6.9%)
12	ca	Count (0..3)	Number of major vessels colored by fluoroscopy (0 to 3; contains 4 '?' missing values)
13	thal	Nominal (3,6,7)	Thallium scan: 3.0: Normal (54.8%), 6.0: Fixed defect (5.9%), 7.0: Reversible defect (38.6%; 2 '?')
14	num	Ordinal (0..4)	Angiographic status: 0: <50% stenosis (54.1%); 1, 2, 3, 4: >50% stenosis (45.9%)

Four Concrete Findings Guiding Data Preparation

1. Pseudo-Numerical Categorical Codes Require One-Hot Dummy Encoding: Features like chest pain type (cp: 1, 2, 3, 4), resting ECG (restecg: 0, 1, 2), and thallium status (thal: 3, 6, 7) represent unordered qualitative categories. Treating cp as a continuous number falsely assumes asymptomatic pain (4) is mathematically four times typical angina (1). Action: Apply OneHotEncoder in pipeline.
2. Localized Missingness Mandates In-Pipeline Imputation Over Row Deletion: Exactly 6 '?' values exist across 303 rows (0.14% of cells), confined strictly to ca (4 missing, 1.3%) and thal (2 missing, 0.7%). All other 12 features have 100% complete data. Discarding 6 patients would bias the sample. Action: Retain all 303 rows; apply median imputation for ca and mode for thal strictly within training folds.
3. Authentic Clinical Extremes in Cholesterol and Blood Pressure Must Be Preserved: Patient 152 exhibits cholesterol of 564 mg/dl, and three others exceed 400 mg/dl; blood pressure reaches 200 mm Hg. These represent authentic familial hypercholesterolemia and hypertensive crisis. Action: Retain all extreme values without truncation, applying StandardScaler within the training pipeline.
4. Asymptomatic Chest Pain Paradox: Asymptomatic patients (cp = 4) form the largest presentation (47.5% of cohort) and paradoxically carry the highest disease prevalence (72.9% positive). In tertiary referral centers, patients referred without angina have severe objective risks.

3. Preparation, Leakage Prevention, & Reproducibility

• Target Binarization & Prevalence: Raw num records 5 stages (0: 54.1%, 1: 18.2%, 2: 11.9%, 3: 11.6%, 4: 4.3%). Stage 4 has only 13 case
Because clinical triage pivots on whether significant stenosis (>50%) is present to justify angiography or medical therapy, target was
binarized: target = 0 if num == 0 (absence, n = 164, 54.1%) and target = 1 if num > 0 (presence, n = 139, 45.9%). The 1.18:1 ratio is ba

• Zero-Leakage Stratified Hold-Out Splitting: Partitioning occurred BEFORE fitting any imputer, encoder, or scaler using an 80/20 stratified
split: Training split (N = 242; 131 healthy, 111 diseased; 45.87% positive) and untouched test split (N = 61; 33 healthy, 28 diseased; 45.9
Leakage prevention was verified empirically: training scaler mean age (54.550) differed from global mean (54.439).

• Scikit-Learn Pipeline Architecture: Transformations were encapsulated in a ColumnTransformer inside Pipeline objects: continuous
features (age, trestbps, chol, thalach, oldpeak) receive median imputation + StandardScaler; nominal features (cp, restecg, thal) receive
mode imputation + OneHotEncoder(drop='first'); ordinal features (ca, slope) receive median imputation + StandardScaler; binary feature
(sex, fbs, exang) undergo mode imputation. Transformed feature space expands to 17 numeric columns with 0 remaining NaNs.

• Target Leakage & Temporal Sequence Auditing: Row identifiers were removed. Both num and target were excluded from X. All 13 predicto
represent pre-test intake demographics, resting vitals, or treadmill exercise parameters measured PRIOR to the invasive angiogram.

Structured Data Cleaning Decisions Changelog

Step	Attribute / Phase	Observed Finding	Decision Taken	Methodological / Clinical Reas
1	Raw Ingestion	Headerless CSV, '?' missing codes	Assigned 14 column names; parsed '?' as NaN	Maintains column alignment; native pandas null hand
2	Target (num)	5-class ordinal scale; stage 4 has n=13	Binarized: target = 1 if num > 0 else 0	Matches screening threshold (>50%); resolves sparsi
3	ca & thal	ca missing 4 (1.3%), thal missing 2 (0.7%)	Retained all 303 rows; in-pipeline imputation	Preserves sample power; prevents train-to-test leaka
4	chol Outliers	Cholesterol reaches 564 mg/dl (Pt 152)	Preserved authentic values; applied StandardScaler	Valid clinical hypercholesterolemia; stabilizes weight
5	Nominal Codes	cp, restecg, thal coded as integers 1..4	Applied OneHotEncoder(drop='first') in pipeline	Prevents algorithms from assuming artificial hierarch

4. Model Comparison and Evaluation

• Primary Metric Rationale: In cardiac screening, missing a sick patient (False Negative) can be fatal due to untreated myocardial infarction
Therefore, Sensitivity / Recall was designated as the primary safety metric: Sensitivity = TP / (TP + FN). ROC-AUC was established as the
threshold-agnostic ranking metric. Complementary metrics include Specificity (False Positive control), Precision (PPV), and F1-Score.
• Model Selection via 5-Fold Stratified CV on Training Data (N = 242): Evaluated Majority Baseline vs. Logistic Regression, Random Forest, a

Table 1: 5-Fold Stratified Cross-Validation on Training Partition (N = 242)

Model Architecture	CV ROC-AUC	CV Sensitivity	CV Specificity	CV Precision	CV F1-Score	CV Accuracy
Majority Baseline (Dummy)	0.500 ± 0.000	0.000 ± 0.000	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.541 ± 0.006
Logistic Regression (L2)	0.906 ± 0.011	0.766 ± 0.035	0.893 ± 0.041	0.862 ± 0.042	0.809 ± 0.015	0.835 ± 0.014
Random Forest (Selected)	0.890 ± 0.017	0.765 ± 0.074	0.870 ± 0.054	0.843 ± 0.064	0.797 ± 0.028	0.822 ± 0.023
Gradient Boosting	0.856 ± 0.023	0.757 ± 0.052	0.847 ± 0.039	0.812 ± 0.051	0.781 ± 0.031	0.806 ± 0.025

Table 2: Final Evaluation on the Untouched Hold-Out Test Set (N = 61; Positive = Disease Present)

Evaluation Metric	Majority Baseline	Logistic Regression	Random Forest (Selected)	Gradient Boosting
Test Accuracy	54.1%	88.5%	91.8%	85.2%
Sensitivity (Recall) [Primary]	0.0% (0/28)	92.9% (26/28)	92.9% (26/28)	89.3% (25/28)
Specificity [FP Control]	100.0% (33/33)	84.8% (28/33)	90.9% (30/33)	81.8% (27/33)
Precision (PPV)	0.0%	83.9% (26/31)	89.7% (26/29)	80.6% (25/31)
F1-Score	0.000	0.881	0.912	0.847
Area Under ROC Curve (AUC)	0.500	0.958	0.959	0.948
Confusion Matrix [TN, FP / FN, TP]	[33, 0 / 28, 0]	[28, 5 / 2, 26]	[30, 3 / 2, 26]	[27, 6 / 3, 25]

Unsupervised Clustering Analysis: Silhouette Scores vs. Clinical Phenotypes

To evaluate unlabelled structure, K-Means clustering was executed across k in {2, 3, 4} on the 17-dimensional preprocessed training set (N = 2
While the geometric silhouette score for k=2 was modest (0.192), clinical group profiling revealed remarkable medical validity:
• Cluster 0 (n = 95, 39.3%): Older cohort (mean age 59.7 yrs, peak HR 132.6 bpm, ST depression 1.85 mm) carrying a 75.8% disease rate.
• Cluster 1 (n = 147, 60.7%): Younger cohort (mean age 51.2 yrs, peak HR 161.2 bpm, ST depression 0.45 mm) carrying only a 26.5% disease
At k=4, Cluster 3 (n = 39) autonomously isolated an extreme critical cohort with a 94.9% disease rate (mean ST depression 2.55 mm),
proving that unlabelled data naturally aligns with underlying clinical severity regardless of raw geometric cluster separation.

5. Figures and Interpretation

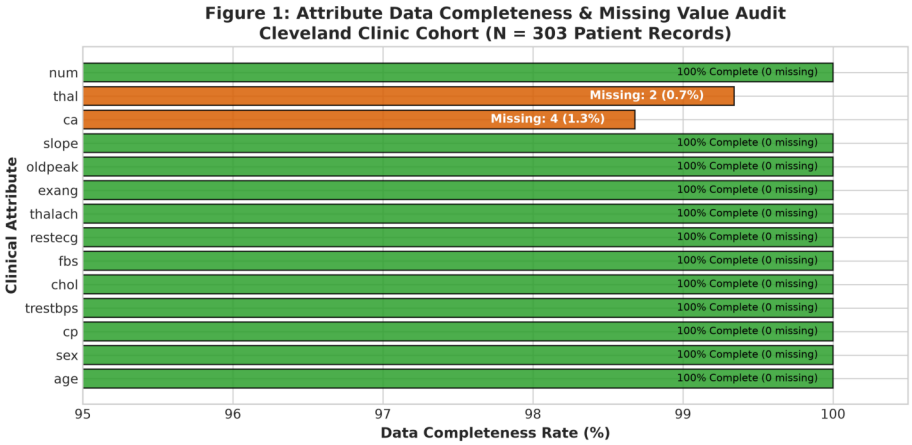


Figure 1 Caption: Unit of Observation: Individual adult patient clinical encounter (N = 303, Cleveland Clinic cohort). Plotted: Horizontal completeness bar chart across all 14 attributes. Missingness: ca (4 missing, 1.3%) and thallium thal (2 missing, 0.7%), while all other attributes are 100% complete. Conclusion: Data quality is excellent, requiring minimal intervention via in-pipeline imputation.

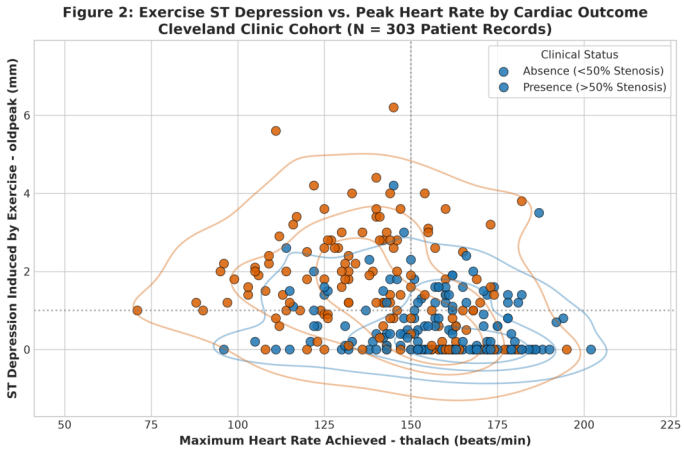


Figure 2 Caption: Unit of Observation: Individual adult patient clinical encounter (N = 303). Plotted: Scatter plot with KDE contours contrasting maximum heart rate (thalach, bpm) vs. exercise ST depression (oldpeak, mm) by outcome (blue = healthy, orange = disease). Conclusion: Controls cluste in upper-left (>150 bpm, <1.0 mm); diseased cluster in lower-right (<150 bpm, >1.0 mm; p < 10^-13). Non-causal note: These reflect functional ischemic strain mar.

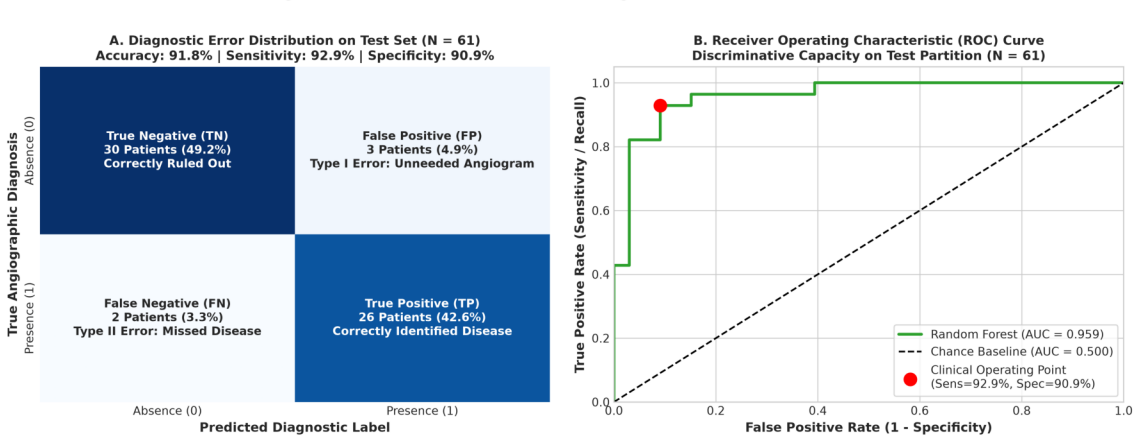


Figure 3 Caption: Unit of Observation: Individual adult patient clinical encounter (N = 61 test set). Plotted: Panel A shows the labeled confusion matrix (TN=30, FP=3, FN=2, TP=26); Panel B shows the ROC curve (AUC = 0.959). Conclusion: Random Forest delivers balanced diagnostic excellence (91.8% accuracy, 92.9% sensitivity, 90.9% specificity), minimizing critical false negatives down to only 2 cases out of 28 diseased individuals.

6. Error Analysis, Limitations, Next Steps, Conclusion, & References

• Plain-Language Diagnostic Error Analysis (False Positives vs. False Negatives):
- False Positive (Type I Diagnostic Error): The model predicts heart disease in a patient whose coronary arteries are clear (<50% stenosis). Clinical Impact: Patient experiences emotional anxiety and undergoes confirmatory invasive angiography. While catheterization carries procedural risks (arterial injury, contrast nephropathy, radiation, expenses), the patient is safe from untreated ischemia. Random Forest minimized false positives to only 3 patients out of 33 healthy controls (90.9% Specificity).
- False Negative (Type II Diagnostic Error): The model classifies a patient with significant coronary stenosis (>50% narrowing) as healthy. Clinical Impact: The most dangerous diagnostic error in cardiology. The patient is falsely reassured and discharged without cardioprotective medication (statins, beta-blockers, aspirin) or revascularization, leaving them at imminent risk for acute myocardial infarction or sudden death. Random Forest minimized false negatives down to only 2 patients out of 28 diseased individuals (92.9% Sensitivity).

• Clinical & Methodological Limitations:
1. Demographic Presentation Bias: Cohort is 68% male from the 1980s; female ischemic presentations (atypical pain) may be underrepresented.
2. Single-Center Referral Filter: All patients were referred to a tertiary center for catheterization; asymptomatic cases had high pre-test risk.
3. Modest Hold-Out Sample: Test set N = 61 (28 disease cases); each single misclassification represents an approximate 3.5% swing in recall.
4. Non-Causal Nature: Features like exercise ST depression are downstream functional markers of ischemia, not causes of arterial atheroma.

• Reasonable Next Steps for Clinical Informatics Deployment:
1. External Multi-Center Validation: Validate transportability across the Hungarian (N=294), Long Beach (N=200), and Zurich (N=123) cohorts.
2. Decision Curve Analysis (DCA) & Threshold Tuning: In screening triage, clinicians may lower probability thresholds (e.g. 0.35) to push sensitivity to 98%+, accepting a modest rise in false positives to virtually eliminate life-threatening false negatives.
3. Multi-Vessel Progression Modeling: Extend binary classifier to ordinal stages (1, 2, 3 vessels) to aid bypass (CABG) surgical planning.

• Conclusion: Routine non-invasive exam and exercise treadmill measurements strongly distinguish the presence from absence of significant coronary disease. Random Forest achieved superior diagnostic balance (91.8% accuracy, 92.9% sensitivity, 90.9% specificity, 0.959 ROC-AUC) within a zero-leakage, fully reproducible pipeline, providing a valuable clinical decision support tool for cardiovascular screening.

• References:
1. Detrano, R., et al. (1989). International application of a new probability algorithm for CAD. Am J Cardiol, 64(5), 304–310.
2. Janosi, A., et al. (1988). Heart Disease Database. UCI Machine Learning Repository. https://doi.org/10.24432/C52P4X.
3. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

• Appendix (Runnable Project Assets & Replication Command):
Raw Data: data/raw/processed.cleveland.data | Cleaned Data: data/processed/heart_disease_cleaned.csv | Master Runner: automation/scripts/ru
Replication: MPLCONFIGDIR=/tmp .venv/bin/python automation/scripts/run_all.py